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COMMUNICATION

Web accessibility evaluation of top-ranking university Web sites in Spain, Chile and Mexico

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Abstract
The purpose of this study is to evaluate the web accessibility of home pages of universities in Spain, Chile and Mexico. The sample used in this study was the 15 top universities listed in the ranking Webometrics. The study has been carried out using automatic tools in order to analyze the level of compliance with the WCAG 2.0 guidelines. The results reveal that the selected Web sites have barriers of access to information and that these difficulties are even greater in the case of those who make use of assistive technology. Users of the analyzed Web sites can encounter difficulties mainly with the perception of information and with navigating the site, just as previous studies of web accessibility have shown. This paper concludes that the 3 analyzed countries show low levels of web accessibility on the selected Web sites, proving that there still exist many shortcomings with regard to meeting the regulation and respecting the fundamental rights of all people.

Keywords Web accessibility · WCAG 2.0 · Disability · University · Evaluation tools

1 Introduction

In the last years, education has undergone a great change. Without a shade of doubt, the development of Information and Communication Technologies (ICT) has revolutionized the processes of teaching and learning and currently almost all universities offer online courses. Universities have evolved in parallel with the new technologies, and they have benefited immensely from the expansion of the Internet and the proliferation of last-generation technological devices. As a consequence of this change, it seems almost strange nowadays to attend a university course without the virtual component. E-learning defined by Seale [1] as "any technology that might support and enhance learning" has managed to make higher education more accessible for a greater number of people [2]. This way of teaching turns out to be in alignment with one of the mission statements established by the United Nations Educational, Scientific and Cultural Organization (UNESCO). Its aim is to "ensure inclusive and equitable quality education and promote lifelong learning opportunities for all" [3]. Nonetheless, it must be taken into account that not all individuals interact with online content the same way; the elderly or people with disabilities sometimes make use of assistive technology. For this reason, it is fundamental for the online learning platforms to be designed using the best practice design principles, with the aim of making the information more accessible for users [4]. In the light of the above, higher education institutions should not only promote learning, education and training of adults and vulnerable groups [5], but they also have a duty to ensure equality of access to educational opportunities. As stated in article 26 of the Universal Declaration of Human Rights "Everyone has the right to education … higher education shall be equally accessible to all on the basis of merit" [6]. The accessibility requirements for an online platform have been established by the World Wide Web Consortium (W3C) which defines accessibility as "universal Access to the Internet, regardless of the kind of hardware, software, network infrastructure, native language, culture, geographical location, or physical or mental ability" [7]. With the objective of protecting the rights of all people, W3C created the Web Accessibility Initiative (WAI). It is a team in charge of establishing guidelines of accessibility for online content known as Web Content Accessibility Guidelines (WCAG). Their goal is to guide web design toward one which would be more universal, reducing the existing barriers and attempting to make the contents accessible to the highest number of people possible

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[8]. WCAG 2.0 is divided into 4 fundamental principles: Perceivable, Operable, Understandable and Robust. These principles are governed by guidelines which are divided into verification criteria categorized into 3 conformance levels: A, AA, and AAA (priority 1, 2 and 3) according to their level of importance and accessibility requirements [9]. Priority 1 and 2 errors seriously affect the use of the web by users with disabilities, and they should be reviewed and corrected in order to meet the current regulation. The objective of this study is to analyze and compare to what degree the requirements of WCAG 2.0 have been met by the homepages of educational platforms of Spanish, Chilean and Mexican universities.

2 Related literature

Numerous studies on web accessibility of online university platforms or e-learning platforms have been carried out by researchers. A great majority of these research studies reveals that universities encounter serious difficulties in meeting the regulation and that greater effort needs to be made to achieve better quality and more user-friendly access to educational content. In order to conduct this type of studies, it is possible to make use of software tools which help to evaluate the degree of accessibility of a Web site [10, 20]. These types of tools are frequently used by researchers to carry out web accessibility studies. For instance, Laitano [11] analyzed to what degree Web sites of Argentinian universities complied with WCAG. This was done using various automatic evaluation tools such as Achecker or Examinator. The study brought to light the fact that none of the Web sites had even the most elemental level of conformance set by WCAG 2.0. Kurt [12] carried out an assessment of the level of accessibility of Turkish university homepages, highlighting the difficulties of access to the online content. Espadinha et al. [13] analyzed homepages of all Public Portuguese universities homepages employing two different automatic assessment tools, Examinator and Hera, which conducted periodic evaluations along the course of 3 years. The authors concluded that the institutions showed a level of accessibility which was by no means acceptable. Ringlaben et al. [14] assessed the accessibility of homepages of 51 University Departments of Special Education, and the study revealed that most of the evaluated pages had accessibility problems. Bringing the focus to the studies on accessibility of higher education institutions in Spain, Chacón-Medina et al. [15] checked the accessibility level of 74 university Web sites with the help of Web Accessibility Test (TAW) [16]. After having verified the principles, guidelines and conformance criteria of priority level A and AA, it was discovered that none of the universities had fully met the aforementioned regulation. That same year, Spanish researchers conducted a comparative study between Spanish universities and those from abroad, the result of which was that the former showed low levels of web accessibility [17]. With regard to the accessibility studies of higher education institutions in Chile, the study which bears most similarity to ours is the one conducted by Vargas et al. [18]. The researchers analyzed 58 university Web sites, evaluating 5 pages of each. The study was carried out using TAW, and its results revealed that none of the Web sites met the minimum requirements established by the W3C. A study which would compare the compliance with the WCAG 2.0 of educational entities in Spain, Chile and Mexico does not exist as of date.

3 Material and method

In order to compare the compliance of higher education institutions in Spain, Chile and Mexico with WCAG 2.0, the first 15 higher education institutions of each country listed in the ranking Webometrics were used as the sample [19]. The said ranking measures the online presence and visibility of universities. Table 1 lists the institutions which were used in the present study.

3.1 Tool selection

Firstly, WAVE [24] was employed in order to evaluate the color combinations of the foreground and the background of the visual presentation of the university Web sites. This tool has previously been used in web accessibility studies of higher education institutions [23, 25–27]. Subsequently, TAW was used to carry out an analysis of the selected webpages according to the WCAG 2.0, in the same way that other researchers have done in the past [17, 18, 28, 29]. This tool provides a detailed report of errors and warnings found on each page, organizing all this information according to the four fundamental principles of web accessibility (Perceptible, Operable, Understandable and Robust). The data obtained using the two automated tools were analyzed according to its distribution. A nonparametric, statistical hypothesis test was then carried out, which is the appropriate measure to compare the statistical significance of the differences found in the set of the three countries, as well as those found in every pair.

3.2 Sample and the selection of webpages to be assessed

The homepage of each institution was subjected to analysis, given that it is the most representative page and the one to which web designers and content developers pay most attention. It is a webpage which is frequently analyzed in web accessibility studies [11, 12, 21, 22, 30, 31].

Table 1 Top 15 higher education centers in Spain, Chile and México, according to the Webometrics ranking

Spain	Chile	Mexico
U. de Barcelona	U. de Chile	U. Nacional Autónoma de México
U. Complutense de Madrid Pontificia	U. Católica de Chile	CINVESTAV
U. de Granada	U. de Concepción	U. de Colima
U. de València	U. Técnica Federico Santa María	U. Autónoma Metropolitana
U. Autònoma de Barcelona	U. de Santiago de Chile	U. de Guadalajara
U. Autónoma de Madrid	U. Austral de Chile Benemérita	U. Autónoma de Puebla
U. Politècnica de Catalunya	U. de Antofagasta	U. Autónoma de Nuevo León
U. Politécnica de Valencia	U. Diego Portales	U. Autónoma del Estado de México
U. Politécnica de Madrid	Pontificia U. Católica de Valparaíso	U. Autónoma de Sinaloa
U. Pompeu Fabra	U. de Valparaíso	U. Michoacana de San Nicolás de Hidalgo
U. de Sevilla	U. de Talca	U. Veracruzana
U. de Zaragoza	U. de Tarapacá	U. de Guanajuato
U. del País Vasco	U. de la Frontera	U. Autónoma de Querétaro
U. de Santiago de Compostela	U. Católica del Norte	U. de las Américas Puebla
U. de Murcia	U. Adolfo Ibáñez	U. Autónoma de Yucatán

Table 2 Contrast errors by country

Country	Mean	Min–Max	Median	Skewness	Kurtosis
Spain	24 (23.57)	0–69	12	.930	− .50
Chile	19 (19.66)	0–69	10	1.35	1.63
Mexico	20.73 (19.99)	1–63	13	.91	− .34

4 Results

4.1 WAVE

WAVE was used to assess the color combinations of the foreground and the background of the visual presentation of the selected Web sites (Table 2). WAVE detects color contrast errors which generate barriers of access to information for people with visual impairments. The analysis shows that the analyzed entities, despite being in 3 different counties, display a similar mean and median. The three distributions are asymmetrical with Chile showing the smallest number of errors (median = 10 and average = 19) followed by Mexico (median = 13 and average = 20.73) and Spain (median = 12 and average = 24). The differences observed in the set of these three countries do not reach statistical significance in the Kruskal–Wallis test ($p = .85$), and they also do not differ statistically in the pairwise comparison using the Mann–Whitney U Test (Spain–Chile, $p = .60$; Spain–Mexico, $p = .81$; Chile–Mexico $p = .74$) (Fig. 1).

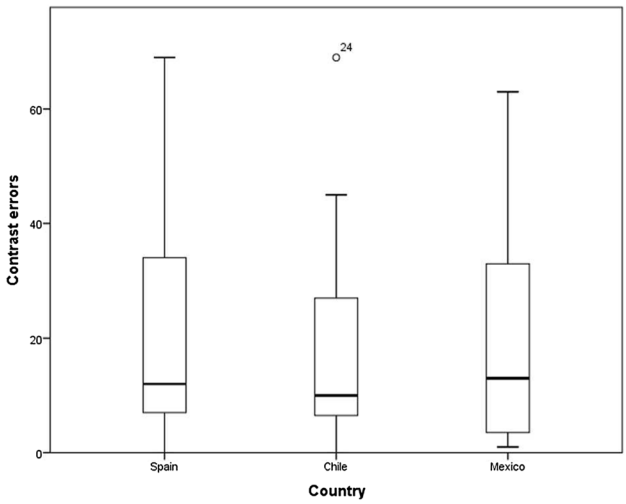


Fig. 1 Contrast errors by country

Table 3 Total errors by country

Country	Mean	Min–Max	Median	Skewness	Kurtosis
Spain	35 (43.84)	4–151	17	2.148	3.75
Chile	85.6 (176.85)	2–718	40	3.73	14.22
Mexico	49.73 (43.02)	2–153	40	1.269	1.22

4.2 TAW

The analysis provided by the TAW tool detects errors related to technical accessibility requirements of conformance level A which have not been met by the three counties (Table 3).

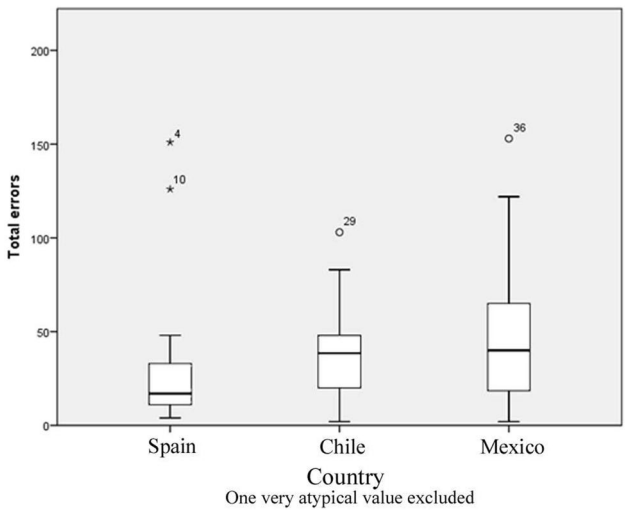


Fig. 2 Distribution of the data

It is clear from the data that there is a great variability among the analyzed Web sites of the three countries. The ranges between minimum and maximum values are very pronounced: 147 in the case of Spain and 716 in the case of Chile, with the latter university showing 718 errors. The kurtosis values indicate that none of the countries have a normal distribution. The three distributions are asymmetrical showing a presence of extreme values, diverging far from the average, especially in the case of Chile and Spain. In the box and whiskers plot (Fig. 2), it can be seen how the atypical errors affect the distribution of the data.

Spain has two extreme values, Chile has an extraordinarily extreme and atypical value, Mexico only has one atypical value. Under these conditions, the optimal measure to express the central tendency is the median rather than the average. The three countries are far from reaching Level A accessibility certification, with Spain showing the smallest number of errors (median = 17 and average = 35), followed by Mexico (median = 40 and average = 49.73) and Chile (median = 40 and average = 85.6). The differences observed in the set of these three countries do not reach statistical significance in the Kruskal–Wallis test ($p = .17$). They also do not differ statistically in the pairwise comparison using the Mann–Whitney *U* Test (Spain–Chile, which only reaches a central tendency, $p = 0.06$; Spain–Mexico, $p = 17$; Chile–Mexico $p = .97$). The results of the study show that all the analyzed entities fail to meet the necessary requirements to reach Level A accessibility certification (Table 4).

Having categorized the errors by principle and country, it can be observed how the Perceptible principle is the one which displays the highest number of errors regardless of the country (Fig. 3).

To carry out an analysis of the results, the accessibility errors found in each country were categorized by principle (Fig. 4). The principle with the biggest amount of errors was the Perceptible principle (45%), making it essential to implement the necessary changes in the structures and the coding of the analyzed webpages. The error most detected by TAW was the omission of textual alternatives to non-textual contents (mostly images), which is of utmost importance to a person who uses a screen reader. Moreover, the tool

Table 4 Accessibility checkpoints violated in Web sites by country

WCAG 2.0 guidelines	Spain	Chile	Mexico
1 Perceivable			
1.1 Text alternatives	103	222	310
1.3 Adaptable	161	331	44
2 Operable			
2.4 Navigable	121	448	255
3 Understandable			
3.1 Readable	2	5	5
3.2 Predictable	4	4	6
3.3 Input Assistance	13	45	11
4 Robust			
4.1 Compatible	140	229	115
Total	544	1284	746

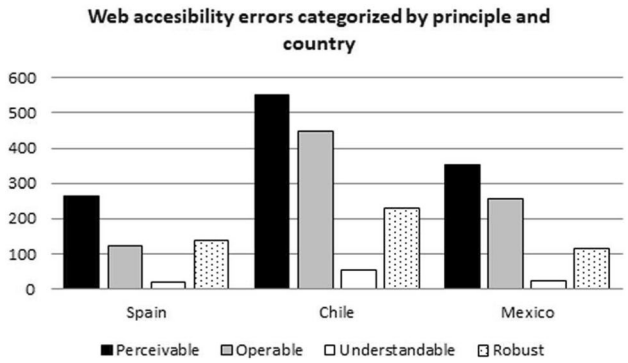


Fig. 3 Web accessibility errors categorized by principle and country

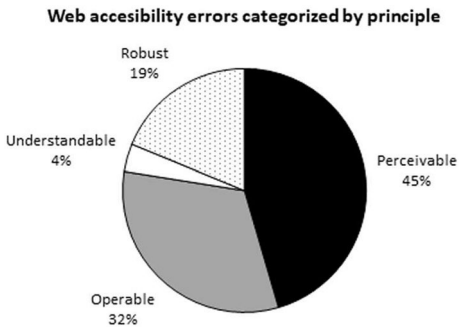


Fig. 4 Web accessibility errors categorized by principle

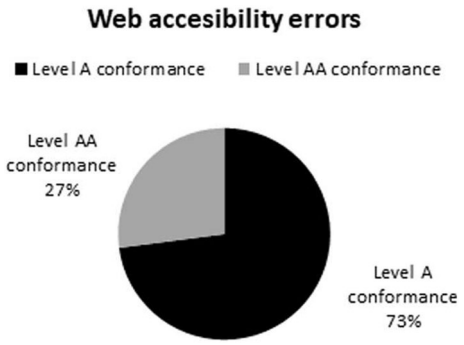


Fig. 5 Web accessibility errors categorized by priority

also detects errors regarding content adaptation according to the needs of each user, which is fundamental to a person who makes use of assistive technology. With regard to the Operable principle (32%), the most detected error resides in the omission of texts in links, which helps users know where they are and toward where they are going to navigate. In terms of the Understandable principle (4%), the most repeated error was the inadequate use of the label element in the forms. Finally, regarding the Robust principle, TAW has found errors in the HTML code which makes up the Web sites. Having categorized the accessibility errors found in the three aforementioned countries, it has been discovered that 73% of the errors belong to conformance level A, while 27% of the same belong to conformance level AA (Fig. 5).

5 Conclusion

The main objective of this work has been to analyze, through the use of automated tools, the degree to which the top 15 Spanish, Chilean and Mexican higher education centers listed in the Webometrics ranking, meet the WCAG 2.0 criteria. The results show that the chosen institutions have difficulties complying with the WCAG 2.0, mostly regarding conformance level A, which is the most basic one. The differences observed among these three countries do not reach statistical significance. This applies to both the set of three countries as well as the sets of pairs, and only Spain shows a slight advantage with regard to the statistical significance of the tendency. The results bear resemblance to previous research studies [11, 15, 18, 32] in which the assessed institutions display precarious levels of web accessibility. Having analyzed the data according to the fundamental principles of the WCAG 2.0 regulation, it can be seen that the Perceptible principle is the one to which web designers and developers ought to pay the most attention. The most common error resides in the omission of textual alternatives, coinciding with [14, 30] as it is the easiest error to solve. Textual alternatives are of great importance to a blind person or one with serious visual impairment since the users interact with the web content through screen readers, computer programs capable of transforming the textual content into voice. In addition, providing the online contents with an adequate color contrast is necessary for all users, especially users with low vision or individuals who cannot differentiate between certain colors [33]. The study reveals that users have difficulties with the use of selected Web sites, especially when it comes to people with disabilities using assistive technology. The use of HTML source code validators designed by W3C could reduce the number of errors, thus facilitating the interaction between assistive technology and the source code of the evaluated Web sites. WCAG 2.0 guidelines have been designed in order to preserve the fundamental rights of people, and the present study reveals difficulties of access to information for the elderly and people with disabilities. The higher educational institutions in Chile, Spain and Mexico ought to increase their efforts to facilitate the accessibility and usability of information technologies by anyone at any place and at any time [34], thus managing to make e-learning much more inclusive.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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